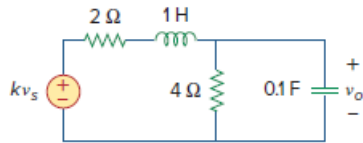


Problem

For the circuit in Fig. 16.95, find $H(s) = V_o(s)/V_s(s)$. Assume zero initial conditions.

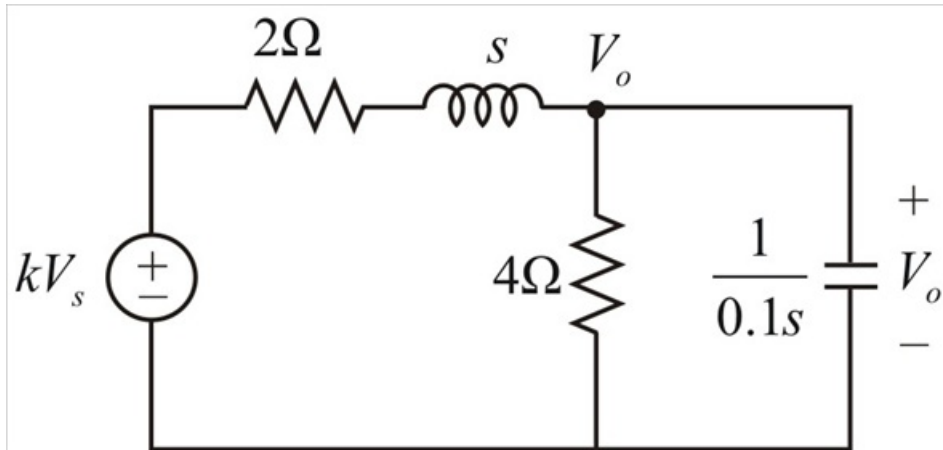
Figure 16.95:



Step-by-step solution

Step 1 of 2

The circuit in s – domain is assuming zero initial conditions



Step 2 of 2

Using Nodal analysis, we get

$$\frac{V_o - kV_s}{s + 2} + \frac{V_o}{4} + \frac{V_o}{\frac{10}{s}} = 0$$

$$V_o \left(\frac{1}{s + 2} + \frac{1}{4} + \frac{s}{10} \right) = \frac{kV_s}{s + 2}$$

$$V_o(s) \left[\frac{40 + 10(s + 2) + 4s(s + 2)}{40(s + 2)} \right] = \frac{kV_s}{(s + 2)}$$

$$\frac{V_o(s)}{V_s(s)} = \frac{40k}{40 + 10s + 20 + 4s^2 + 8s}$$

$$\frac{V_o(s)}{V_s(s)} = \frac{40k}{4s^2 + 18s + 60}$$

$$\therefore H(s) = \frac{20k}{2s^2 + 9s + 30}$$